

**UNIVERSITY OF HEALTH
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TACHYCARDIA

Tachycardia

1. Definition

Tachycardia is defined as a resting heart rate greater than 100 beats per minute (bpm) in adults, originating from the sinoatrial (SA) node, the atrioventricular (AV) junction, or the ventricles (Al-Khatib et al., 2018; Page et al., 2016). It may be:

- Physiological, when it represents an appropriate response to increased metabolic demands (e.g., exercise, pain, anxiety, fever, pregnancy).
- Pathological, when it is disproportionate to physiological needs or caused by intrinsic abnormalities of cardiac impulse formation or conduction.

From an electrophysiological standpoint, tachycardia includes several major categories:

- Sinus tachycardia: Originating from the sinoatrial node, with normal P waves and PR interval, but an elevated rate (>100 bpm).
- Supraventricular tachycardia (SVT): Tachycardias arising above the ventricles, such as atrioventricular nodal re-entrant tachycardia (AVNRT), atrioventricular re-entrant tachycardia (AVRT), and focal atrial tachycardia.
- Atrial fibrillation (AF) and atrial flutter with rapid ventricular response (RVR): Irregular or regular atrial tachyarrhythmias associated with fast ventricular rates.
- Ventricular tachycardia (VT): A rapid rhythm originating in the ventricles, usually wide-complex, and frequently life-threatening (Al-Khatib et al., 2018).

For nursing practice, it is essential to distinguish benign sinus tachycardia from potentially life-threatening tachyarrhythmias that require urgent intervention.

2. Incidence and Epidemiology

The incidence and prevalence of tachycardia vary according to the specific arrhythmia type.

- **Sinus tachycardia** is extremely common in both hospital and community settings, as it frequently represents a normal response to stress, illness, or exertion. The exact prevalence is difficult to quantify due to its transient nature.
- **Inappropriate sinus tachycardia (IST)**, characterised by persistently elevated sinus rate at rest and disproportionate response to minimal exertion, has an estimated prevalence of 1.2–1.6% and is more frequently reported in young women (Olshansky & Sullivan, 2019).
- **Supraventricular tachycardia (SVT)** has an incidence of approximately 35 per 100,000 person-years and a prevalence of around 2.3 per 1,000 persons, with rising frequency in older adults (Page et al., 2016).
- **Atrial fibrillation (AF)**, often presenting as tachycardia, has a global prevalence estimated at 2–4% in adults, increasing sharply with age and comorbidities such as hypertension, diabetes, and heart failure (Hindricks et al., 2021).
- **Ventricular tachycardia (VT)** occurs mainly in individuals with structural heart disease (e.g., post–myocardial infarction scar, cardiomyopathy). Although less common than Supraventricular tachycardia and Atrial fibrillation, its incidence increases with age and is a major cause of sudden cardiac death (Priori et al., 2015).

Key epidemiological trends include:

- ✓ Population ageing, leading to more **Atrial fibrillation** and **Ventricular tachycardia** Rising prevalence of cardiovascular risk factors (obesity, diabetes, hypertension).
- ✓ Increased use of stimulants and recreational drugs in some populations.
- ✓ Nurses must be aware of these trends to anticipate patient risk and contribute to early detection and prevention.

3. Causes (Etiology)

Tachycardia results from a wide spectrum of cardiac and non-cardiac causes. Identification of the underlying aetiology is critical to effective nursing and medical management.

3.1 Physiological and Compensatory Causes

- a) Exercise and physical exertion
- b) Emotional stress, anxiety, and pain
- c) Fever and systemic infection
- d) Pregnancy and physiological hypervolaemia
- e) Hypovolaemia due to dehydration, haemorrhage, or fluid loss
- f) Shock states (e.g., sepsis, anaphylaxis, cardiogenic shock) where tachycardia is a compensatory response to maintain cardiac output

These are usually sinus tachycardias that resolve with correction of the precipitating factor.

3.2 Cardiac Causes

- i. Ischaemic heart disease and acute coronary syndromes
- ii. Heart failure and various cardiomyopathies
- iii. Valvular heart diseases (e.g., mitral stenosis, mitral regurgitation)
- iv. Congenital heart disease
- v. Post-myocardial infarction scar tissue providing substrate for re-entrant Ventricular tachycardia
- vi. Conduction system abnormalities (e.g., Wolff–Parkinson–White [WPW] syndrome, atrioventricular nodal re-entrant tachycardia, atrioventricular re-entrant tachycardia)
- vii. Atrial fibrillation and atrial flutter (often with rapid ventricular response)

3.3 Non-Cardiac Pathological Causes

- a. Anaemia (reduced oxygen-carrying capacity)
- b. Hyperthyroidism and other endocrine disorders (e.g., pheochromocytoma)
- c. Hypoxia due to respiratory diseases (e.g., chronic obstructive pulmonary disease, pneumonia, pulmonary embolism)
- d. Metabolic derangements (acidosis, hypoglycaemia)
- e. Electrolyte disturbances (hypokalaemia, hypomagnesaemia, hypocalcaemia)
- f. Autonomic dysfunction conditions, including postural orthostatic tachycardia syndrome (POTS)
- g. Severe pain or distress

3.4 Iatrogenic and Substance-Related Causes

Medications:

- Beta-agonists (e.g., salbutamol)
- Decongestants (e.g., pseudoephedrine)
- Thyroxine excess
- Anticholinergic drugs
- Some antidepressants and antipsychotics (especially those that prolong QT interval)
- Digoxin toxicity (can cause atrial or ventricular tachyarrhythmias)

Substances:

- Caffeine and other energy drinks
- Nicotine
- Cocaine, amphetamines, and other sympathomimetic illicit drugs
- Acute and chronic alcohol intake, and alcohol withdrawal

Nursing assessment should systematically screen for these factors to guide diagnostic and therapeutic decisions.

4. Pathophysiology

The pathophysiology of tachycardia involves abnormalities in impulse formation and/or conduction within the cardiac conduction system.

4.1 Enhanced Automaticity

Pacemaker cells in the sinoatrial (SA) node or ectopic foci in atrial, junctional, or ventricular tissue may develop increased automaticity, meaning they depolarise more rapidly than normal. Contributing factors include:

- Increased sympathetic tone and catecholamine levels
- Hyperthyroidism
- Fever
- Ischaemia
- Certain drugs (e.g., catecholamines, theophylline)

This mechanism underlies sinus tachycardia and some forms of atrial and ventricular tachycardia (Al-Khatib et al., 2018).

4.2 Triggered Activity

Triggered activity occurs when early or delayed after depolarisations reach threshold potential and initiate extra beats or tachycardia. Common triggers include:

- Electrolyte imbalances (e.g., hypokalaemia, hypomagnesaemia)

- Digoxin toxicity
- Myocardial ischaemia or reperfusion
- Proarrhythmic drug effects (e.g., QT-prolonging agents)

Triggered activity may cause focal atrial tachycardia and certain forms of polymorphic Ventricular tachycardia (VT) (Priori et al., 2015).

4.3 Re-Entry Circuits

Re-entry is a major mechanism for many clinical tachycardias. It requires:

- ➔ Two pathways with different conduction velocities and refractory periods
- ➔ A unidirectional block in one pathway
- ➔ A trigger (premature beat) to initiate circular conduction

Once initiated, the impulse travels in a circuit, repeatedly depolarising cardiac tissue. Re-entry explains:

- ✓ Atrioventricular nodal re-entrant tachycardia AVNRT and atrioventricular re-entrant tachycardia AVRT (typical **Supraventricular tachycardia s**)
- ✓ Typical atrial flutter
- ✓ Many monomorphic Ventricular tachycardia VTs in structural heart disease (scar tissue)

4.4 Haemodynamic Consequences

Sustained tachycardia can impair cardiovascular function by:

- ✓ Reduced diastolic filling time: Less time for ventricular filling, reducing stroke volume.
- ✓ Decreased coronary perfusion: Coronary blood flow predominantly occurs in diastole; fast heart rates reduce myocardial oxygen supply.

- ✓ Increased myocardial oxygen demand: Higher rate and contractility require more oxygen.

As a result, tachycardia may lead to:

- Hypotension and poor organ perfusion
- Worsening myocardial ischaemia
- Precipitation or deterioration of heart failure
- Tachycardia-induced cardiomyopathy if persistent (Hendricks et al., 2021)

5. Clinical Features (Signs and Symptoms)

The clinical manifestations of tachycardia depend on the underlying rhythm, heart rate, duration of the episode, and the presence of structural heart disease or other comorbidities.

5.1 Common Symptoms

- Palpitations (awareness of a rapid, pounding, or irregular heartbeat)
- Chest discomfort or chest pain
- Shortness of breath, especially on exertion
- Dizziness or light-headedness
- Presyncope or syncope (fainting), particularly with **Ventricular tachycardia** VT or very rapid **Supraventricular tachycardia** SVT
- Fatigue and exercise intolerance
- Anxiety and sense of impending doom

5.2 Physical Signs

Resting heart rate greater than 100 bpm (apical pulse)

- i. Irregular rhythm in Atrial fibrillation AF
- ii. Hypotension or labile blood pressure
- iii. Tachypnoea and signs of respiratory distress
- iv. Cool, clammy skin and diaphoresis in unstable patients
- v. Signs of poor perfusion (prolonged capillary refill, reduced urine output)
- vi. Evidence of heart failure: raised jugular venous pressure, peripheral oedema, basal crackles on lung auscultation.
- vii. Altered mental status (confusion, agitation, reduced consciousness) in severe cases.

For nurses, a key clinical task is to distinguish haemodynamically stable tachycardia from haemodynamically unstable tachycardia, as instability (hypotension, shock, ongoing chest pain, acute heart failure, altered consciousness) indicates the need for immediate medical intervention (Page et al., 2016).

6. Diagnostic Investigations

Nurses play a crucial role in initiating, preparing for, and monitoring diagnostic investigations.

6.1 Bedside Assessment

- Full set of vital signs (HR, BP, RR, temperature, SpO₂)
- Rapid cardiovascular and respiratory examination
- Assessment of level of consciousness using validated scales
- Pain assessment (e.g., chest pain characteristics)
- Perfusion assessment: skin colour and temperature, capillary refill, urine output

6.2 Electrocardiographic Evaluation

- i. 12-lead ECG: The primary tool to identify arrhythmia type (sinus tachycardia, Supraventricular tachycardia SVT, Atrial fibrillation AF, flutter, Ventricular tachycardia) by providing information on rate, rhythm, P-wave morphology, QRS width, and ST-T changes (Page et al., 2016).
- ii. Continuous cardiac monitoring/telemetry: Allows detection of paroxysmal episodes, rate variability, and arrhythmia burden.
- iii. Ambulatory monitoring: Holter monitoring (24–48 hours) or event recorders for intermittent symptoms in outpatient settings.

Nurses should be competent in obtaining high-quality ECGs and recognising gross abnormalities such as very fast rates, wide-complex rhythms, or marked ST-T changes that require urgent review.

6.3 Laboratory Studies

- Full blood count to identify anaemia or infection.
- Serum electrolytes (potassium, magnesium, calcium), urea and creatinine
- Thyroid function tests (TSH, free T4, free T3) to detect hyperthyroidism.
- Cardiac biomarkers (e.g., troponin) if myocardial ischaemia is suspected.
- B-type natriuretic peptide (BNP or NT-proBNP) in suspected heart failure
- Arterial blood gases in hypoxia or critically ill patients
- Drug levels (e.g., digoxin) when toxicity is suspected.

6.4 Imaging and Specialised Tests

- Echocardiography: Assesses chamber size and function, wall motion, valvular abnormalities, and presence of structural heart disease.
- Chest X-ray: Provides information regarding heart size and pulmonary congestion and helps identify respiratory causes of dyspnoea.
- Coronary angiography or CT coronary angiography: Performed in patients with suspected obstructive coronary artery disease.
- Electrophysiology (EP) studies: Invasive mapping of cardiac electrical pathways; often used to diagnose complex tachyarrhythmias and to guide catheter ablation (Priori et al., 2015).

7. Medical Management

Management strategies focus on stabilising the patient, treating reversible causes, controlling heart rate or restoring normal rhythm, and preventing recurrence and complications.

7.1 Initial Assessment and Stabilisation

- Airway, breathing, circulation (ABC) assessment and support
- Oxygen therapy if hypoxic ($\text{SpO}_2 < 94\%$) or in respiratory distress
- Establishment of intravenous access and continuous ECG and pulse oximetry monitoring
- Rapid identification and management of reversible factors (e.g., fluid resuscitation for hypovolaemia, analgesia for severe pain, antipyretics for fever, treatment of sepsis)

7.2 Pharmacological Management

The choice of drug depends on the type of tachycardia and haemodynamic status.

- Beta-blockers (e.g., metoprolol, bisoprolol, esmolol):

- Reduce heart rate, myocardial oxygen consumption, and sympathetic tone.
- Used for sinus tachycardia (when appropriate), Supraventricular tachycardia SVT, and Atrial fibrillation AF with rapid ventricular response.
- Must be used cautiously in asthma, severe COPD, bradycardia, and decompensated heart failure (Al-Khatib et al., 2018).
- Non-dihydropyridine calcium channel blockers (e.g., diltiazem, verapamil):
- Slow atrioventricular AV nodal conduction and are useful in rate control of Supraventricular tachycardia SVT and Atrial fibrillation AF.
- Avoid in some Ventricular tachycardia VTs and in severe systolic heart failure.

Adenosine:

- A short-acting atrioventricular AV nodal blocking agent used acutely for diagnosis and termination of certain Supraventricular tachycardia SVTs.
- Administered as a rapid IV bolus under continuous monitoring due to risk of transient asystole and bronchospasm (Page et al., 2016).
- Antiarrhythmic drugs (e.g., amiodarone, procainamide, flecainide in selected patients):
- Used in ventricular tachycardia or in atrial arrhythmias when first-line measures are ineffective or contraindicated.
- Require careful monitoring for hypotension, QT prolongation, and organ toxicity.

Digoxin:

- ➔ Slows atrioventricular AV nodal conduction and may help control ventricular rate in Atrial fibrillation AF, especially in heart failure.
- ➔ Has a narrow therapeutic index; toxicity can itself precipitate tachyarrhythmias.

Rate and rhythm control in Atrial fibrillation AF:

- ❖ Rate control with beta-blockers, diltiazem/verapamil, or digoxin.
- ❖ Rhythm control with pharmacological (e.g., amiodarone) or electrical cardioversion in selected patients.

Anticoagulation (Atrial fibrillation AF and atrial flutter):

Oral anticoagulants (warfarin or direct oral anticoagulants) to reduce stroke risk, guided by CHA₂DS₂-VASc and bleeding risk scores (Hindricks et al., 2021).

Aetiology-directed therapy:

Antibiotics for sepsis, antithyroid medications for hyperthyroidism, blood transfusion for severe anaemia, and correction of electrolyte disturbances.

7.3 Non-Pharmacological Management.

Vagal manoeuvres:

- ❖ Modified Valsalva manoeuvre, carotid sinus massage (only when no carotid disease is present).
- ❖ Increase vagal tone and may terminate AV node–dependent Supraventricular tachycardia SVTs (Page et al., 2016).

Synchronized cardioversion:

Indicated for haemodynamically unstable tachycardias or for stable tachyarrhythmias refractory to medications.

Used for Supraventricular tachycardia SVT, Atrial fibrillation AF/flutter, and Ventricular tachycardia VT with a pulse.

Requires appropriate sedation and fully equipped resuscitation facilities.

Defibrillation:

Used for pulseless Ventricular tachycardia VT or ventricular fibrillation as part of advanced life support.

Catheter ablation:

- Definitive non-pharmacological treatment for many Supraventricular tachycardia SVTs, typical atrial flutter, some Atrial fibrillation AF, and certain Ventricular tachycardia VTs.
- Reduces arrhythmia burden and may allow discontinuation of antiarrhythmics (Priori et al., 2015).

Implantable cardioverter-defibrillator (ICD):

Indicated for high-risk patients with sustained Ventricular tachycardia or those at high risk of sudden cardiac death.

8. Nursing Care and Interventions

Nursing care is fundamental to the safe and effective management of patients with tachycardia.

8.1 Assessment and Monitoring

- a. Vital signs: Monitor heart rate, blood pressure, respiratory rate, oxygen saturation, and temperature at appropriate intervals based on acuity.
- b. ECG monitoring: Continuously observe rhythm, rate trends, and changes in ECG morphology.

- c. Perfusion status: Assess skin colour and temperature, capillary refill, peripheral pulses, and urine output.
- d. Respiratory status: Evaluate work of breathing, auscultate lungs, and monitor for signs of pulmonary oedema.
- e. Neurological status: Assess level of consciousness and orientation to detect cerebral hypoperfusion.

8.2 Acute Nursing Interventions

- a. Position the patient in semi-Fowler's to improve comfort and ventilation.
- b. Administer oxygen as prescribed and adjust based on saturation and clinical status.
- c. Establish and maintain intravenous access; ensure patency for medication and fluid administration.
- d. Assist in performing vagal manoeuvres when prescribed, ensuring continuous monitoring and patient safety.
- e. Prepare the patient and equipment for potential cardioversion or defibrillation:
- f. Confirm identity and procedure, check consent if feasible.
- g. Attach defibrillator pads correctly and ensure all team members are aware of "shock" phases.
- h. Provide post-procedure monitoring of vital signs, rhythm, and level of consciousness.

8.3 Medication Management and Safety

- a. Administer prescribed antiarrhythmics and rate-control medications accurately, verifying the "five rights" (right patient, drug, dose, time, and route).

- b. Monitor for adverse drug reactions such as bradycardia, hypotension, proarrhythmia, bronchospasm, and organ toxicity.
- c. Check relevant laboratory values (e.g., electrolytes, drug levels, renal function) before and during therapy.
- d. Educate patients about the purpose and schedule of medications and emphasise adherence.

8.4 Patient Education

Effective patient education includes:

- **Explanation of condition:** Provide clear, jargon-free information about the type of tachycardia, likely causes, and treatment plan.
- **Symptom recognition:** Teach patients and caregivers to recognise warning signs such as severe palpitations, chest pain, dyspnoea, and presyncope/syncope, and when to seek emergency help.
- **Self-monitoring:** Instruct appropriate patients on how to check pulse rate and rhythm if recommended.

Lifestyle modifications:

Smoking cessation

- Limiting or avoiding caffeine, alcohol, and energy drinks
- Avoiding illicit drugs and non-prescribed stimulants
- Maintaining a healthy weight and engaging in moderate physical activity as advised
- Employing stress-reduction techniques such as relaxation, breathing exercises, or mindfulness (Hindricks et al., 2021)

Medication adherence and safety:

- Stress the importance of not stopping beta-blockers or antiarrhythmics abruptly.
- Explain anticoagulant precautions, including bleeding risk, drug and food interactions, and the need for regular monitoring where applicable.

8.5 Psychosocial Support

- Acknowledge that symptoms such as palpitations and syncope can be frightening and lead to anxiety or depression.
- Provide emotional support, encourage questions, and promote shared decision-making.
- Refer to psychological support services or support groups when appropriate.

8.6 Interprofessional Collaboration

- a. Communicate promptly with physicians and the cardiology team regarding changes in patient condition or ECG findings.
- b. Collaborate with pharmacists on medication selection, dosing, and interactions.
- c. Work with physiotherapists, dietitians, and rehabilitation specialists to implement comprehensive care and secondary prevention strategies.

9. Prevention

Prevention of tachycardia focuses on reducing risk factors, controlling underlying disease processes, and avoiding precipitating triggers.

9.1 Primary Prevention

- Promotion of a heart-healthy lifestyle, including a balanced diet, regular physical activity, weight management, and smoking cessation.
- Early detection and management of hypertension, diabetes, dyslipidaemia, and sleep apnoea.
- Avoidance or limitation of stimulant substances such as caffeine and nicotine.

9.2 Secondary Prevention

- Ongoing management of established cardiovascular disease to prevent arrhythmic complications.
- Regular follow-up of patients with known arrhythmias to monitor symptom control and treatment effectiveness.
- Timely treatment of acute triggers such as infections, pain, and electrolyte disturbances.

9.3 Tertiary Prevention

- Adherence to antiarrhythmic drugs and anticoagulants to prevent recurrence and thromboembolic events.
- Use of catheter ablation or ICDs in indicated patients to reduce the risk of recurrent Ventricular tachycardia VT and sudden cardiac death.
- Participation in cardiac rehabilitation programmes to optimise physical and psychological recovery (Anderson et al., 2016).

Nurses contribute to prevention by engaging in health education, screening, and ongoing support.

10. Rehabilitation

Rehabilitation supports patients in regaining functional capacity and quality of life after arrhythmic events or related cardiac interventions.

Cardiac rehabilitation programmes:

- Structured exercise training tailored to individual capacity and cardiac status.
- Education on disease mechanisms, medication management, and lifestyle modification.
- Psychological support to address anxiety, depression, and fear of recurrence (Anderson et al., 2016).

Activity guidance:

Nurses provide advice on gradual increase in activity, recognising symptoms that require rest or medical review (e.g., excessive dyspnoea, chest pain, marked palpitations).

Long-term follow-up:

- Monitoring symptom patterns, adherence to therapies, and risk factor control in outpatient or community settings.
- Coordination with general practitioners and cardiologists for ongoing care.

Specific support for ICD recipients:

- Education about device function, what to do after a shock, and safety issues (e.g., driving, work restrictions according to local guidelines).
- Emotional support regarding fear and adjustment to living with an implanted device.

11. Complications

Uncontrolled or recurrent tachycardia can lead to serious and potentially fatal complications.

Haemodynamic compromise and shock:

Sustained rapid rates can severely reduce cardiac output, resulting in hypotension, organ hypoperfusion, and multi-organ dysfunction.

Myocardial ischaemia and infarction:

Elevated heart rate increases oxygen demand while reducing coronary perfusion time, aggravating ischaemia, especially in coronary artery disease.

Heart failure and tachycardia-induced cardiomyopathy:

Persistent tachycardia can cause ventricular remodelling and systolic dysfunction, sometimes reversible with adequate rate or rhythm control (Hindricks et al., 2021).

Thromboembolism and stroke:

In Atrial fibrillation AF and atrial flutter, blood stasis in the atria promotes thrombus formation, which can embolise to cerebral, systemic, or peripheral arteries.

Sudden cardiac death:

Sustained Ventricular tachycardia VT can degenerate into ventricular fibrillation or asystole, leading to cardiac arrest if not treated promptly.

Medication-related complications:

Proarrhythmic effects, bradycardia, hypotension, organ toxicity (e.g., thyroid, hepatic, pulmonary toxicity with chronic amiodarone use).

Nursing vigilance and timely escalation of care are essential to minimise these adverse outcomes.

12. Conclusion

Tachycardia encompasses a broad range of conditions, from benign physiological responses to life-threatening arrhythmias such as sustained ventricular tachycardia. A thorough understanding of its definition, epidemiology, causes, and pathophysiology enables nurses to perform accurate assessment and contribute to timely recognition of high-risk presentations. Evidence-based management incorporates pharmacological and non-pharmacological strategies, tailored to the specific arrhythmia type and underlying disease processes.

Nursing practice is central to effective tachycardia care. Critical responsibilities include continuous monitoring, safe medication administration, early recognition of deterioration, and support during emergency interventions such as cardioversion and defibrillation. Equally important are patient education, psychosocial support, and promotion of lifestyle changes that reduce the risk of recurrence and complications. By integrating acute management with prevention and rehabilitation, nurses play a pivotal role in improving clinical outcomes,

reducing hospital readmissions, and enhancing quality of life for patients living with tachycardia.

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